Exercise J-M

1. If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then $\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ is :

(Here arg(z) denotes the principal argument of complex number z) **[JEE(Main)July2021]**

यदि z तथा ω दो सम्मिश्र संख्याएँ है, जिनके लिए $|z\omega| = 1$ तथा $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$ बराबर है, तो $\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ बराबर है : (जहाँ arg(z) सम्मिश्र संख्या z के मुख्य कोणांक को दर्शाता है) **[JEE(Main)July2021]**

(A) $-\frac{\pi}{4}$ (B) $-\frac{3\pi}{4}$ (C) $\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

Ans. B

Sol. $|z\omega| = 1 = |\bar{z}\omega|$

$\arg(\omega) - \arg(z) = -3\pi/2$

$\arg(\omega\bar{z}) = -3\pi/2$

$\omega\bar{z} = i \qquad (\because (\omega\bar{z}) = 1)$

$$\arg\left(\frac{1-2i}{1+3i}\right) = \arg(1-2i) - \arg(1+3i)$$

$= -\tan^{-1}(2) - \tan^{-1}(3)$

$= -3\pi/4$

2. If α and β are the distinct roots of the equation $x^2 + (3)^{1/4}x + 3^{1/2} = 0$, then the value of $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ is equal to : **[JEE(Main)July2021]**

यदि α तथा β, समीकरण $x^2 + (3)^{1/4}x + 3^{1/2} = 0$ के दो भिन्न मूल है, तो $\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$ का मान बराबर है : **[JEE(Main)July2021]**

(A) 56×3^{25} (B) 56×3^{24} (C) 28×3^{25} (D) 52×3^{24}

Ans. (D)

Sol. $x^2 + (3^{1/4})x + 3^{1/2} = 0$

$\alpha = 3^{1/4}\omega \qquad \beta = 3^{1/4}\omega^2$

$\alpha^{12} = 3^3 = 27 \qquad \beta = 3^3$

$\alpha^{96} = 3^{24} \qquad \beta^{96} = 3^{24}$

$\alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1)$

$= 3^{24}(3^3 - 1) + 3^{24}(3^3 - 1)$

$= 2(26)(3^{24})$

$= 52 \times 3^{24}$

3. Let $A = [a_{ij}]$ be a real matrix of order 3×3, such that $a_{i1} + a_{i2} + a_{i3} = 1$, for $i = 1,2,3$. Then, the sum of all the entries of the matrix A^3 is equal to : **[JEE(Main)July2021]**

माना $A = [a_{ij}]$, कोटि 3×3 का एक वास्तविक आव्यूह इस प्रकार है कि प्रत्येक $i = 1, 2, 3$ के लिए $a_{i1} + a_{i2} + a_{i3} = 1$ है। तो आव्यूह A^3 की सभी प्रविष्टियों का योग बराबर है : **[JEE(Main)July2021]**

(A) 9 (B) 1 (C) 2 (D) 3

Ans. (D)

Sol. $$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Let } X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow \quad AX = X$

Replace X by AX

$A(AX) = AX$

$A^2X = AX = X$

$A^2X = X$

Replace X by AX

$A^3X = AX = X$

$$\text{Let } A^3 \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

Now

$$A^3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \\ z_1 + z_2 + z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

hence sum of elements = 3

4. Let n denote the number of solutions of the equation $z^2 + 3\bar{z} = 0$, where z is a complex number. Then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is equal to : **[JEE(Main)July2021]**

माना सम्मिश्र संख्या z के लिए समीकरण $z^2 + 3\bar{z} = 0$ के मूलों की संख्या n है। तो $\sum_{k=0}^{\infty} \frac{1}{n^k}$ का मान बराबर है :

[JEE(Main)July2021]

(A) 2 (B) 4/3 (C) 1 (D) 3/2

Ans. (B)

Sol. $z^2 + 3\bar{z} = 0$

let $x = x + iy$

$x^2 - y^2 + 2ixy + 3(x - iy) = 0$

By comparision

$x^2 - y^2 + 3x = 0$ ...(1)

$2xy - 3y = 0$ ...(2)

$y(2x - 3) = 0$

$y = 0$

$x = 3/2$

If $y = 0$

then by (1) we get $x = 0, -3$

hence $x = 0, y = 0$ and $x = 0$ and $x = -3$ are two solutions.

If $x = 3/2$

then by (1) we get $y = \pm\frac{\sqrt{27}}{2}$

$x = \frac{3}{2}$, $y = \frac{\sqrt{27}}{2}$

$x = 3/2$, $y = -\frac{\sqrt{27}}{2}$ are two solution

hence total solution $n = 4$

$$\sum_{k=0}^{\infty} \frac{1}{4^k} = 1 + \frac{1}{4} + \frac{1}{4^2} + ... \infty = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

5. Let C be the set of all complex numbers. Let

$S_1 = \{z \in C \mid |z - 3 - 2i|^2 = 8\}$,

$S_2 = \{z \in C \mid Re(z) \geq 5|\}$ and

$S_3 = \{z \in C \mid |z - \bar{z}| \geq 8\}$.

Then the number of elements is $S_1 \cap S_2 \cap S_3$ is equal to : **[JEE(Main)July2021]**

माना सभी सम्मिश्र संख्याओं का समुच्चय C है। माना :

$S_1 = \{z \in C \mid |z - 3 - 2i|^2 = 8\}$,

$S_2 = \{z \in C \mid Re(z) \geq 5|\}$ तथा

$S_3 = \{z \in C \mid |z - \bar{z}| \geq 8\}$हैं।

तो $S_1 \cap S_2 \cap S_3$ में अवयवों की संख्या बराबर है : **[JEE(Main)July2021]**

(A) 1 (B) 0 (C) 2 (D) Infinite (अनन्त)

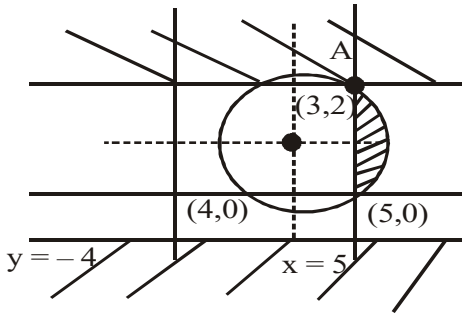
Ans. (A)

Sol. $S_1 : |z = (3 + 2i)| = 2\sqrt{2}$

$S_2 : x \geq 5$

$S_3 : |y| \geq 4$

$S_1 \cap S_2 \cap S_3 = A$



6. Let $\mathbb{C}$ be the set of all complex numbers. Let

$S_1 = \{z \in \mathbb{C} : |z-2| \leq 1\}$ and $S_2 = \{z \in \mathbb{C} : z(1+i) + \overline{z}(1-i) \geq 4\}$.

Then, the maximum value of $\left|z - \frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to : **[JEE(Main)July2021]**

माना सभी सम्मिश्र संख्याओं का समुच्चय $\mathbb{C}$ है।

माना $S_1 = \{z \in \mathbb{C} : |z-2| \leq 1\}$ तथा $S_2 = \{z \in \mathbb{C} : z(1+i) + \overline{z}(1-i) \geq 4\}$ हैं। तो $z \in S_1 \cap S_2$ के लिए $\left|z - \frac{5}{2}\right|^2$ का अधि

िकतम मान बराबर है : **[JEE(Main)July2021]**

(A) $\frac{3+2\sqrt{2}}{4}$ (B) $\frac{5+2\sqrt{2}}{2}$ (C) $\frac{3+2\sqrt{2}}{2}$ (D) $\frac{5+2\sqrt{2}}{4}$

Ans. (D)

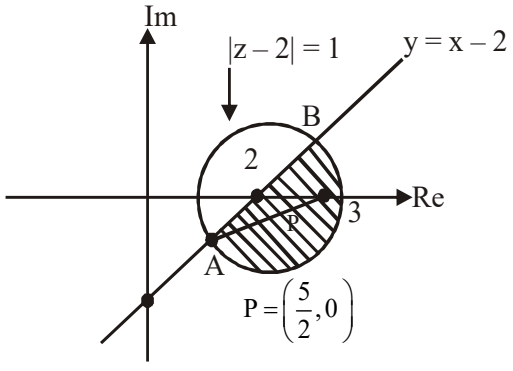
Sol. $S_1 : |z-2| \leq |$ _____(i)

$S_2 : z(1+i) + \overline{z}(1-i) \geq 4$

$(x+iy)(1+i) + (x-iy)(1-i) \geq 4$

$x - y \geq 2$

$y \leq x - 2$ _____(ii)



Equation of circle $(x-2)^2 + y^2 = 1$ _____(iii)

for A & B $y^2 + y^2 = 1$

$2y^2 = 1$

$$y = \pm \frac{1}{\sqrt{2}}$$

$$A = \left(2 - \frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$$

maximum value of $\left|z - \frac{5}{2}\right| = (PA)^2$

$$= \left|2 - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i - \frac{5}{2}\right|^2$$

$$= \left|\frac{-1}{2} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i\right|^2 = \frac{5+\sqrt{2}}{4}$$

7. Let $S = \left\{ n \in N \middle| \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right\}$, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is _____ . **[JEE(Main)July2021]**

यदि $S = \left\{ n \in N \middle| \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right\}$, जहाँ $i = \sqrt{-1}$ हैं तो समुच्चय S में दो अंकों वाली संख्याओं की संख्या है _____ **[JEE(Main)July2021]**

Ans. 11

Sol. $$S = \left\{ n \in N \middle| \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right.$$

let $A = \begin{bmatrix} 0 & i \\ 1 & 0 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 0 & i \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & i \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$$

$$A^4 = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^8 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$A^{16} = A^{24} = A^{32} ..... = A^{96} = I$

No. of two digits numbers = 11

8. The equation of a circle is $Re(z^2) + 2(Im(z))^2 + 2Re(z) = 0$, where $z = x + iy$. A line which passes through the center of the given circle and the vertex of the parabola, $x^2 - 6x - y + 13 = 0$, has y-intercept equal to _____.

[JEE(Main)July2021]

एक वृत का समीकरण $Re(z^2) + 2(Im(z))^2 + 2Re(z) = 0$ है, जबकि $z = x + iy$ है, तो एक रेखा, जो इस वृत के केन्द्र से तथा परवलय $x^2 - 6x - y + 13 = 0$ के शीर्ष से होकर जाती है, का y-अक्ष पर अन्तः खण्ड है _____ **[JEE(Main)July2021]**

Ans. 1

Sol. $Z = x + iy$

$Z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy$

$Re(Z^2) = x^2 - y^2 \quad Im(z) = y$

$x^2 - y^2 + 2y^2 + 2x = 0$

$x^2 + y^2 + 2x = 0$

centre $(-1, 0)$

$x^2 - 6x = y - 13$

$(x - 3)^2 - 9 = y - 13$

$(x - 3)^2 = y - 4$

$x - 3 = X \quad y - 4 = Y$

$X^2 = Y$

Vertex $= (3, 4)$

$$\frac{4-0}{4} = \frac{y-0}{x+1}$$

$$1 = \frac{y}{x+1}$$

$n + 1 = y$

Put $x = 0$

yint $= 1$

9. If the real part of the complex number $z = \frac{3 + 2i\cos\theta}{1 - 3i\cos\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos\theta$ is equal to_____.

[JEE(Main)July2021]

यदि सम्मिश्र संख्या $z = \frac{3 + 2i\cos\theta}{1 - 3i\cos\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ का वास्तविक भाग शून्य हैं, तो $\sin^2 3\theta + \cos^2\theta$ का मान बराबर है _____

[JEE(Main)July2021]

Ans. 1

Sol. $$z = \frac{3 + 2i\cos\theta}{1 - 3i\cos\theta}$$

$$z = \frac{(3 + 2i + \cos\theta)(1 + 3i + \cos\theta)}{1 + 2\cos^2\theta}$$

$$\text{Real part of } z = \frac{3-6\cos^2\theta}{1+9\cos^2\theta} = 0$$

$$\therefore \cos^2\theta = \frac{1}{2} \qquad \text{_____(1)}$$

$\sin^2 3\theta + \cos^2\theta$

$= (3\sin^2\theta - 4\sin^3\theta)^2 + \cos^2\theta$

$= \sin^2\theta\,(3 - 4\sin^2\theta)^2 + \cos\theta$

$$= \left(1-\frac{1}{2}\right)\left(4\times\frac{1}{2}-1\right)^2 + \frac{1}{2}$$

$$= \frac{1}{2}(2-1)^2 + \frac{1}{2}$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

10. The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with : **[JEE(Main)Aug.2021]**

(A) centre at (0, 0) and radius $\sqrt{2}$ (B) centre at (0, – 1) and radius $\sqrt{2}$

(C) centre at (0, 1) and radius 2 (D) centre at (0, 1) and radius $\sqrt{2}$

समीकरण $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ एक वृत्त को निरूपित करता है जिसका : **[JEE(Main)Aug.2021]**

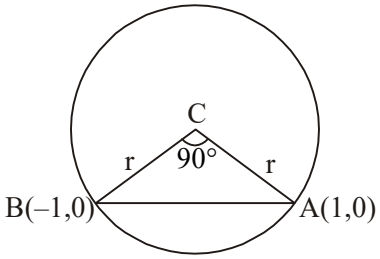
(A) केन्द्र (0, 0) है तथा त्रिज्या $\sqrt{2}$ है (B) केन्द्र (0, – 1) है तथा त्रिज्या $\sqrt{2}$

(C) केन्द्र (0, 1) है तथा त्रिज्या 2 है (D) केन्द्र (0, 1) है तथा त्रिज्या $\sqrt{2}$ है

Ans. (D)

Sol. $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

Chord AB will subtend 90° at center.



$2r^2 = (AB)^2 \Rightarrow \quad r = \sqrt{2}$

Center = (0, 1)

11. If $\left(\sqrt{3}+i\right)^{100} = 2^{99}\ (p+iq)$, then p and q are roots of the equation : **[JEE(Main)Aug.2021]**

यदि $\left(\sqrt{3}+i\right)^{100} = 2^{99}\ (p+iq)$ है, तो p तथा q किस समीकरण के मूल है? **[JEE(Main)Aug.2021]**

(A) $x^2+\left(\sqrt{3}-1\right)x-\sqrt{3}=0$ (B) $x^2+\left(\sqrt{3}+1\right)x+\sqrt{3}=0$

(C) $x^2-\left(\sqrt{3}+1\right)x+\sqrt{3}=0$ (D) $x^2-\left(\sqrt{3}-1\right)x-\sqrt{3}=0$

Ans. (D)

Sol. $(\sqrt{3}+i)^{100} = 2^{99}(p+iq)$

$$2^{100}\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)^{100} = 2^{99}(p+iq)$$

$$2.e^{i\frac{50\pi}{3}} = p+iq$$

$$2\left(\cos\frac{50\pi}{3}+i\sin\frac{50\pi}{3}\right) = p+iq$$

$$2\left(\frac{-1}{2}+i\frac{\sqrt{3}}{2}\right) = p+iq$$

$p=-1$ and $q=\sqrt{3}$

$x^2-(\sqrt{3}-1)x-\sqrt{3}=0$

12. If $S=\left\{z\in\mathbb{C}:\frac{z-i}{z+2i}\in\mathbb{R}\right\}$, then : **[JEE(Main)Aug.2021]**

(A) S contains exactly two elements (B) S is a circle in the complex plane

(C) S is a straight line in the complex plane (D) S contains only one element

यदि $S=\left\{z\in\mathbb{C}:\frac{z-i}{z+2i}\in\mathbb{R}\right\}$ है, तो : **[JEE(Main)Aug.2021]**

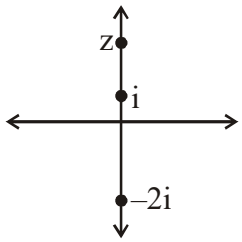
(A) S में ठीक दो अवयव है (B) S सम्मिश्र समतल में, एक वृत्त है

(C) S सम्मिश्र समतल में, एक सरल है (D) S में केवल एक अवयव है

Ans. (C)

Sol. $\frac{z-i}{z-(-2i)}\in R$

$$\arg\left(\frac{z-i}{z-(-2i)}\right)=0 \text{ or } \pi$$



13. If $a_r = \cos\frac{2r\pi}{9} + i\sin\frac{2r\pi}{9}$, $r = 1, 2, 3, ..., i = \sqrt{-1}$, then the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is equal to :

[JEE(Main)Aug.2021]

यदि $a_r = \cos\frac{2r\pi}{9} + i\sin\frac{2r\pi}{9}$, $r = 1, 2, 3, ..., i = \sqrt{-1}$, तो सारणिक $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ बराबर है :

[JEE(Main)Aug.2021]

(A) $a_2a_6 - a_4a_8$ (B) $a_1a_9 - a_3a_7$ (C) a_9 (D) a_5

Ans. (B)

Sol. $a_r = e^{i\frac{2r\pi}{9}}$, Let $\frac{2\pi}{9} = \theta \Rightarrow a_r = e^{ir\theta}$

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} e^{i\theta} & e^{i2\theta} & e^{i3\theta} \\ e^{i4\theta} & e^{i5\theta} & e^{i6\theta} \\ e^{i7\theta} & e^{i8\theta} & e^{i9\theta} \end{vmatrix}$$

$C_2 \to C_2 - C_1$

$$= \begin{vmatrix} e^{i\theta} & e^{i\theta}(e^{i\theta}-1) & e^{i3\theta} \\ e^{i4\theta} & e^{i4\theta}(e^{i\theta}-1) & e^{i6\theta} \\ e^{i7\theta} & e^{i4\theta}(e^{i\theta}-1) & e^{i9\theta} \end{vmatrix} = 0$$

Now $a_1a_9 - a_3a_7 = e^{i\theta}.e^{i9\theta} - e^{i3\theta}.e^{i7\theta}$

$= 0$

14. If z is a complex number such that $\frac{z-i}{z-1}$ is purely, imaginary, then the minimum value of $|z-(3+3i)|$ is :

[JEE(Main)Aug.2021]

माना z एक सम्मिश्र संख्या है, जिसके लिए $\frac{z-i}{z-1}$ पूर्ण रूप से काल्पनिक है, तो $|z-(3+3i)|$ का निम्नतम मान है :

[JEE(Main)Aug.2021]

(A) $3\sqrt{2}$ (B) $6\sqrt{2}$ (C) $2\sqrt{2}$ (D) $2\sqrt{2}-1$

Ans. (C)

Sol. $z = x + iy$

$$\frac{x+iy-i}{x+iy-1} = \frac{x+i(y-1)}{x-1+iy} \times \frac{(x-1)-iy}{(x-1)-iy}$$

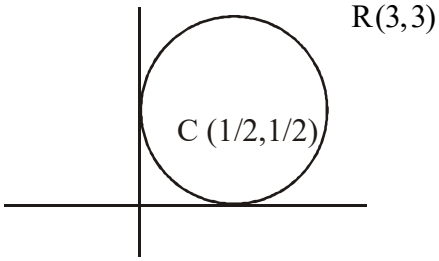
$$= \frac{x(x-1)-ixy+i(y-1)(x-1)+y(y-1)}{(x-1)^2+y^2}$$

$$= \frac{x^2-x+y^2-y+i((y-1)(x-1)-xy)}{(x-1)^2+y^2}$$

$$\frac{x^2+y^2-x-y}{(x-1)^2+y^2} = 0$$

$$x^2+y^2-x-y=0$$

$$C=\left(\frac{1}{2},\frac{1}{2}\right) \qquad r = \frac{1}{\sqrt{2}}$$



$$|z-(3+3i)|_{min} = CR - r$$

$$= \sqrt{\left(\frac{5}{2}\right)^2+\left(\frac{5}{2}\right)^2} - \frac{1}{\sqrt{2}}$$

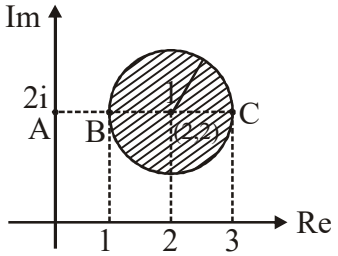
$$= \frac{5}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

15. If for the complex numbers z satisfying $|z-2-2i| \le 1$, the maximum value of $|3iz+6|$ is attained at $a+ib$, then $a+b$ is equal to _____. **[JEE(Main)Aug.2021]**

यदि $|z-2-2i| \le 1$ को संतुष्ट करने वाली सम्मिश्र संख्याओं z के लिए $|3iz+6|$ का उच्चतम मान $a+ib$ पर प्राप्त होता है, तब $a+b$ के बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 5

Sol.



$$|z-2-zi| \le 1$$

$$\Rightarrow |z-(2-zi)| \le 1$$

It represents circuler region including circumference whose centre is (2, 2) and radius is 1.

$|3iz+6|=|(3i)(z-2i)|$

$=3|z-2i|$

For maximum value of $|3iz+6|$, the value $|z-2i|$ should be maximum and it is maximum, when z lies at C.

$z=3+2i$

$a=3,\ b=2$

$a+b=5$

16. A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{4}$, then the minimum value of $\left|z-9\sqrt{2}-2i\right|^2$ is equal to _____. **[JEE(Main)Aug.2021]**

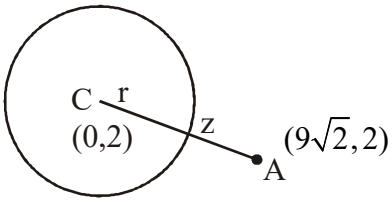
एक बिन्दु z सम्मिश्र समतल में इस प्रकार चलायमान है कि $\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{4}$ है, तो $\left|z-9\sqrt{2}-2i\right|^2$ का न्यूनतम मान बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 98

Sol. Let $z=x+iy$

$$\frac{z-2}{z+2}=\frac{(x-2)+iy}{(x+2)+iy}\times\frac{(x+2)-iy}{(x+2)-iy}$$

$$\frac{z-2}{z+2}=\frac{(x^2+y^2-4)+iy4}{x^2+y^2+4x+4}$$



$$\arg\left(\frac{z-2}{z+2}\right)=\frac{\pi}{4}\Rightarrow x^2+y^2-4=4y$$

$\Rightarrow x^2+y^2-4y=4$

$\Rightarrow x^2+(y-2)^2=8\Rightarrow z$ lies on the circle

$|z-(9\sqrt{2}+2i)|^2_{min}=(AC-r)^2$

$=98$

17. Let z_1 and z_2 be two complex members such that $\arg(z_1-z_2)=\frac{\pi}{4}$ and z_1, z_2 satisfy the equation $|z-3|=\text{Re}(z)$. Then the imaginary part of z_1+z_2 is equal to _____. **[JEE(Main)Aug.2021]**

माना z_1 तथा z_2 दो सम्मिश्र संख्यायें इस प्रकार हैं कि $\arg(z_1-z_2)=\frac{\pi}{4}$ है तथा z_1, z_2 समीकरण $|z-3|=\text{Re}(z)$ को संतुष्ट करते है तो z_1+z_2 का काल्पनिक भाग है_____ **[JEE(Main)Aug.2021]**

Ans. 6

Sol. $z_1=x_1+iy_1$

$z_2 = x_1 + iy_2$

$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$

$\arg(z_1 - z_2) = \dfrac{\pi}{4}$

$\tan^{-1}\left(\dfrac{y_1 - y_2}{x_1 - x_2}\right) = \dfrac{\pi}{4}$

$y_1 - y_2 = x_1 - x_2$ _____(1)

Now $|z_1 - z_2| = \text{Re}(z_1)$

$(x_1 - 3)^2 + y_1^2 = x_1^2$ _____(2)

$(x_2 - 3)^2 + y_1^2 = x_2^2$ _____(3)

(B) – (3)

$(x_1 - 3)^2 - (x_2 - 3)^2 + y_1^2 - y_2^2 = (x_1^2 - x_2^2)$

$-6x_1 + 6x_2 + (y_1 + y_2)(y_1 - y_2) = 0$

$-6(x_1 - x_1) + (y_1 + y_2)(y_1 - y_2) = 0$

but $x_1 - x_2 = y_1 - y_2$ by (1) them

$-6(y_1 - y_2) + (y_1 + y_2)(y_1 - y_2) = 0$

$y_1 \neq y_2$ then $y_1 + y_2 = 6$

18. The least positive integer n such that $\dfrac{(2i)^n}{(1-i)^{n-2}}, i = \sqrt{-1}$, is a positive integer, is _____.

[JEE(Main)Aug.2021]

निम्नतम धनात्मक पूर्णांक n जिसके लिए $\dfrac{(2i)^n}{(1-i)^{n-2}}, i = \sqrt{-1}$, एक धनात्मक पूर्णांक है, है_____

[JEE(Main)Aug.2021]

Ans. 6

Sol. $\dfrac{(2i)^n}{(1-i)^{n-2}} = \dfrac{(2i)^n}{(-2i)^{\frac{n-2}{2}}}$

$= \dfrac{(2i)^{\frac{n+2}{2}}}{(-1)^{\frac{n-2}{2}}} = \dfrac{2^{\frac{n+2}{n}; i^{\frac{n+2}{2}}}}{(-1)^{\frac{n-2}{2}}}$

This is positive integer for $n = 6$

19. Let $z=\frac{1-i\sqrt{3}}{2}, i=\sqrt{-1}$. Then the value of $21+\left(z+\frac{1}{z}\right)^3+\left(z^2+\frac{1}{z^2}\right)^3+\left(z^3+\frac{1}{z^3}\right)^3+.....+\left(z^{21}+\frac{1}{z^{21}}\right)^3$ is _____.

[JEE(Main)Aug.2021]

माना $z=\frac{1-i\sqrt{3}}{2}, i=\sqrt{-1}$ है। तो $21+\left(z+\frac{1}{z}\right)^3+\left(z^2+\frac{1}{z^2}\right)^3+\left(z^3+\frac{1}{z^3}\right)^3+.....+\left(z^{21}+\frac{1}{z^{21}}\right)^3$ का मान बराबर है _____

[JEE(Main)Aug.2021]

Ans. 13

Sol. $z=\frac{1-i\sqrt{3}}{2}=e^{-i\frac{\pi}{3}}$

$$z^n+\frac{1}{z^n}=e^{-i\frac{n\pi}{3}}+e^{i\frac{n\pi}{3}}=2\cos\frac{n\pi}{3}$$

$z^n+\frac{1}{z^n}$

→ 2	n = 6K
→ 1	n = 6K+1
→ -1	n = 6K + 2
→ -2	n = 6K + 3
→ -1	n = 6K + 4
→ 1	n = 6K + 5

Sum = $21 + 4(1)^3 + 4(-1)^3 + 4(-2)^3 + 3(-1)^3 + 3(1)^3 + 3(2)^3 = 21 + 1 - 1 - 8 = 13$

20. Let $A=\{z\in C:1\le|z-(1+i)|\le 2\}$ and $B=\{z\in A:|z-(1-i)|=1\}$. Then, B :

[JEE(Main)June2022]

(A) is an empty set

(B) contains exactly two elements

(C) contains exactly three elements

(D) is an infinite set

माना $A=\{z\in C:1\le|z-(1+i)|\le 2\}$ तथा $B=\{z\in A:|z-(1-i)|=1\}$ है, तब, B :

[JEE(Main)June2022]

(A) एक रिक्त समुच्चय है

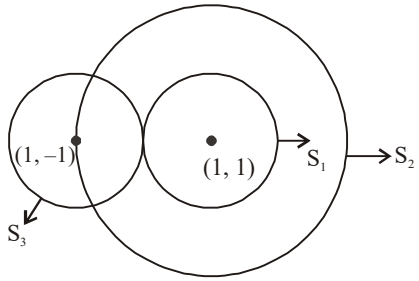
(B) में ठीक दो अवयव हैं

(C) में ठीक तीन अवयव हैं

(D) एक अनन्त समुच्चय है

Ans. D

Sol. Set A ≡ region between two concentric circle, centre is (1, 1)



(1, –1) will lie on onto circle

B ≡ Portion of circle S_3 lies between concentric circle S_1 and S_2

≡ infinite points in B

21. Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then arg(z) is equal to :

[JEE(Main)June2022]

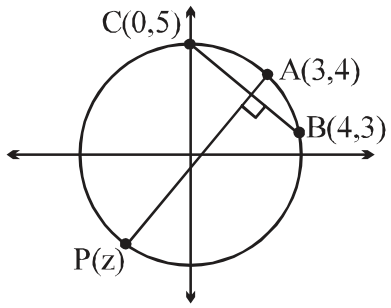
माना सम्मिश्र समतल में एक वृत्त C बिन्दुओं $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ तथा $z_3 = 5i$ से होकर जाता है। यदि वृत्त C पर एक बिन्दु $z(\neq z_1)$ इस प्रकार है कि z तथा z_1 से होकर जाने वाली रेखा z_2 तथा z_3 से होकर जाने वाली रेखा के लंबवत है, तो arg(z) बराबर है :

[JEE(Main)June2022]

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right)-\pi$ (B) $\tan^{-1}\left(\frac{24}{7}\right)-\pi$ (C) $\tan^{-1}(3)-\pi$ (D) $\tan^{-1}\left(\frac{3}{4}\right)-\pi$

Ans. B

Sol.



Slope of $BC = \frac{3-5}{4-0} = -\frac{1}{2}$

Slope of $AP = 2$

equation of $AP : y - 4 = 2(x - 3)$

$\Rightarrow \quad y = 2(x - 1)$

P lies on circle $x^2 + y^2 = 25$

$\Rightarrow x^2 + (2(x-1))^2 = 25$

$\Rightarrow x = -\frac{7}{5}$ and $y = -\frac{24}{5}$

$\Rightarrow \arg(z) = \tan^{-1}\left(\frac{24}{7}\right) - \pi$

22. Let z_1 and z_2 be two complex numbers such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$. Then :

[JEE(Main)June2022]

माना दो सम्मिश्र संख्याओं z_1 तथा z_2 के लिए $\overline{z}_1 = i\overline{z}_2$ तथा $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$ हैं। तो : **[JEE(Main)June2022]**

(A) $\arg z_2 = \frac{\pi}{4}$ (B) $\arg z_2 = -\frac{3\pi}{4}$ (C) $\arg z_1 = \frac{\pi}{4}$ (D) $\arg z_1 = -\frac{3\pi}{4}$

Ans. C

Sol. $z_1 = i\overline{z}_2 \Rightarrow z_1 = -i\overline{z}_2$

$z_2 = iz_1$

$\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi \Rightarrow \quad \arg(z_1) - \arg(\overline{z}_2) = \pi$

$\arg(z_1) + \arg(z_2) = \pi$ _____(i)

$\arg(\overline{z}_1) = \arg(i\overline{z}_2)$

$-\arg(z_1) = \arg(i) + \arg(\overline{z}_2)$

$-\arg(z_1) = \frac{\pi}{2} - \arg(z_2)$

$-\arg(z_1) + \arg(z_2) = \frac{\pi}{2}$ _____(ii)

From (i) – (ii)

$2\arg(z_1) = \frac{\pi}{2}$

$\arg(z_1) = \frac{\pi}{4}$

23. Let $A = \left\{z \in C : \left|\frac{z+1}{z-1}\right| < 1\right\}$ and $B = \left\{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$. Then $A \cap B$ is :

[JEE(Main)June2022]

(A) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(B) a portion of a circle centred at $\left(0,-\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrants only

(C) an empty set

(D) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in third quadrants only

माना $A=\left\{z\in C:\left|\frac{z+1}{z-1}\right|<1\right\}$ तथा $B=\left\{z\in C:\arg\left(\frac{z-1}{z+1}\right)=\frac{2\pi}{3}\right\}$ हैं। तो $A\cap B$:

[JEE(Main)June2022]

(A) केन्द्र $\left(0,-\frac{1}{\sqrt{3}}\right)$ के वृत्त का एक भाग है, जो केवल दूसरे तथा तीसरे चतुर्थांश में है

(B) केन्द्र $\left(0,-\frac{1}{\sqrt{3}}\right)$ के एक वृत्त का एक भाग है जो केवल दूसरे चतुर्थांश में है

(C) एक रिक्त समुच्चय है

(D) त्रिज्या $\frac{2}{\sqrt{3}}$ के एक वृत्त का एक भाग है, जो केवल तीसरे चतुर्थांश में है

Ans. B

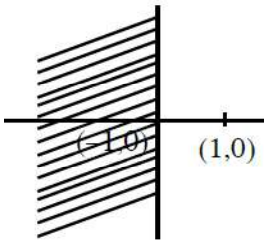
Sol. Set A

$\Rightarrow \left|\frac{z+1}{z-1}\right|<1$

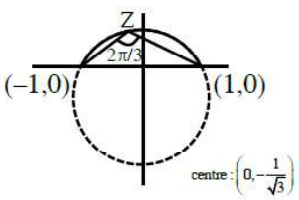
$\Rightarrow |z+1|<|z-1|$

$\Rightarrow (x+1)^2+y^2<(x-1)^2+y^2$

$\Rightarrow x<0$



Set B



$$\Rightarrow \arg\left(\frac{z-1}{z+1}\right)=\frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x-1}\right)-\tan^{-1}\left(\frac{y}{x+1}\right)=\frac{2\pi}{3}$$

$$\Rightarrow x^2+y^2+\frac{2y}{\sqrt{3}}-1=0$$

$A \cap B$

$$\Rightarrow \text{Centre}\left(0,-\frac{1}{\sqrt{3}}\right)$$

24. The area of the polygon, whose vertices are the non-real roots of the equation $\bar{z}=iz^2$ is :

[JEE(Main)June2022]

उस बहुभुज का क्षेत्रफल, जिसके शीर्ष समीकरण $\bar{z}=iz^2$ के अवास्तविक मूल हैं, है : **[JEE(Main)June2022]**

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Ans. A

Sol. $\bar{z}=iz^2$

$|\bar{z}|=|iz^2|$

$|z|=|z^2|$

$|z|(|z|-1)=0$

$|z|=0$ OR $|z|=1$

$z=0$ OR $z\bar{z}=1$

$z\bar{z}=iz^3$

$iz^3=1$

$z^3=-i$

$z^3=e^{i\left(-\frac{\pi}{2}+2k\pi\right)}$

$z=e^{i\left(\frac{-\frac{\pi}{2}+2k\pi}{3}\right)}$

$k=0 \quad z_0=e^{-i\frac{\pi}{6}}=\left(\frac{\sqrt{3}}{2},\frac{-1}{2}\right)$

$k=1 \quad z_1=e^{i\frac{\pi}{2}}=(0,1)$

$k = 2 \quad z_2 = e^{i7\frac{\pi}{6}} = \left(\frac{-\sqrt{3}}{2}, -\frac{1}{2}\right)$

$$\text{Area} = \frac{1}{2}\begin{vmatrix} \sqrt{\frac{3}{2}} & -\frac{1}{2} & 1 \\ 0 & 1 & 1 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 1 \end{vmatrix}$$

$$= \frac{3\sqrt{3}}{4}$$

25. The number of points of intersection of $|z-(4+3i)| = 2$ and $|z| + |z-4| = 6, z \in C$, is :

[JEE(Main)June2022]

$|z-(4+3i)| = 2$ तथा $|z| + |z-4| = 6, z \in C$ के प्रतिच्छेदन बिन्दुओं की संख्या है : **[JEE(Main)June2022]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $|z-(4+3i)| = 2$ is a circle $(x-4)^2 + (y-3)^2 = 2^2$

$|z| + |z-4| = 6$ is an ellipse with focus (0, 0) & (4, 0)

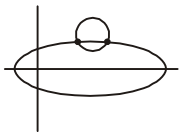
and $2a = 6$ $2ae = 4$ centre (2, 0)

$\Rightarrow a = 3$ $e = \frac{2}{3}$ $\Rightarrow b^2 = 5$

$$\frac{(x-2)^2}{9} + \frac{y^2}{5} = 1$$

Lower extremity of vertical diameter (4, 1)

put in ellipse $S_1 = \frac{(4-2)^2}{9} + \frac{1^2}{5} - 1 = \frac{-16}{45} < 0$

$\Rightarrow$ lies inside ellipse $\Rightarrow$ 

26. Let for some real number α and β, $a = \alpha - i\beta$. If the system of equations $4ix + (1+i)y = 0$ and $8\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)x + \bar{a}y = 0$ has more than one solution then $\frac{\alpha}{\beta}$ is equal to :

[JEE(Main)June2022]

माना α तथा β के किसी वास्तविक मानों के लिए $a = \alpha - i\beta$ है। यदि समीकरण निकाय $4ix + (1 + i)y = 0$ तथा $8\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)x+\bar{a}y=0$ के एक से अधिक हल हैं, तो $\frac{\alpha}{\beta}$ बराबर है : **[JEE(Main)June2022]**

(A) $-2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) $2+\sqrt{3}$ (D) $-2-\sqrt{3}$

Ans. B

Sol. $$\frac{i}{2\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)}=\frac{1+i}{\alpha+i\beta}$$

$$\alpha i+i^2\beta=(1+i)\left(-1+i\sqrt{3}\right)$$

$$-\beta+\alpha i=\left(-1-\sqrt{3}\right)+\left(\sqrt{3}-1\right)i$$

$$\beta=1+\sqrt{3}, \qquad \alpha=\sqrt{3}-1$$

$$\frac{\alpha}{\beta}=\frac{\sqrt{3}-1}{\sqrt{3}+1}=2-\sqrt{3}$$

27. Let α be a root of the equation $1 + x^2 + x^4 = 0$. Then the value of $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ is equal to : **[JEE(Main)June2022]**

माना समीकरण $1 + x^2 + x^4 = 0$ का एक मूल α है। तो $\alpha^{1011} + \alpha^{2022} - \alpha^{3033}$ का मान बराबर है : **[JEE(Main)June2022]**

(A) 1 (B) α (C) $1 + \alpha$ (D) $1 + 2\alpha$

Ans. A

Sol. $1+x^2+x^4 \Rightarrow \left(1+x+x^2\right)\left(1-x+x^2\right)=0$

one root is ω & other is ω^2

$\therefore \omega^3=1 \Rightarrow \alpha^3=1$

then $\alpha^{1011}+\alpha^{2022}-\alpha^{3033}=\left(\alpha^3\right)^{337}+\left(\alpha^3\right)^{674}-\left(\alpha^3\right)^{1011}$

$= 1 + 1 - 1 = 1$

28. Let arg(z) represent the principal argument of the complex number z. Then, $|z| = 3$ and $\arg(z-1)-\arg(z+1)=\frac{\pi}{4}$ intersect : **[JEE(Main)June2022]**

(A) exactly at one point (B) exactly at two points

(C) nowhere (D) at infinitely many points

माना arg(z) सम्मिश्र संख्या z के मुख्य आयाम को निरूपित करता है। तो $|z| = 3$ तथा $\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$:

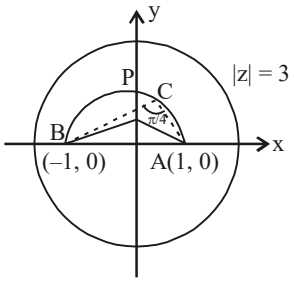
[JEE(Main)June2022]

(A) मात्र एक बिन्दु पर काटते हैं (B) मात्र दो बिन्दुओं पर काटते हैं

(C) किसी भी बिन्दु पर नहीं काटते हैं (D) असंख्य बिन्दुओं पर काटते है

Ans. C

Sol. $|z| = 3, \arg(z-1) - \arg(z+1) = \frac{\pi}{4}$



$\angle AKL = \angle ACB = \frac{\pi}{4}$

$\Rightarrow LK = AL = \alpha = 1$

$\Rightarrow K(0, 1)$

Radius $= \sqrt{2}$

Now $PL = PK + KL = \sqrt{2} + 1$

So, point $P(0, 1 + \sqrt{2})$

$\therefore$ number of intersection $= 0$

29. Let $S = \{z \in C : |z-3| \le 1 \text{ and } z(4+3i) + \bar{z}(4-3i) \le 24\}$. If $\alpha + i\beta$ is the point in S which is closest to 4i, then 25 $(\alpha + \beta)$ is equal to _____. **[JEE(Main)June2022]**

माना $S = \{z \in C : |z-3| \le 1$ तथा $z(4+3i) + \bar{z}(4-3i) \le 24\}$ हैं। यदि S में 4i के निकटतम बिन्दु $\alpha + i\beta$ है, तो 25 $(\alpha + \beta)$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 80

Sol. here $|z-3| \le 1$

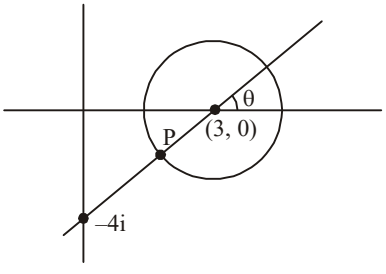
$(x-3)^2 + y^2 \le 1$

and

$z(4+3i) + \bar{z}(4-3i) \le 24$

$4x - 3y \le 12$

$\tan\theta = \frac{4}{3}$



here we need point P

$P(3 - \cos\theta, \sin\theta)$

$P\left(3 - \frac{3}{5}, \frac{4}{5}\right)$

$P\left(\frac{12}{5}, \frac{4}{5}\right)$

$$\alpha + i\beta = \left(\frac{12}{5} + i\frac{4}{5}\right)$$

$$\alpha = \frac{12}{5}$$

$$\beta = \frac{4}{5}$$

$$\alpha + \beta = \frac{16}{5}$$

$25(\alpha + \beta) = 80$

30. If $z^2 + z + 1 = 0, z \in C$, then $\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ is equal to _____. **[JEE(Main)June2022]**

यदि $z^2 + z + 1 = 0, z \in C$ हैं, तो $\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ बराबर है _____ **[JEE(Main)June2022]**

Ans. 2

Sol. If $z^2 + z + 1 = 0, z \in c$

$z = w, w^2$

$$\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$$

$$\left|\sum_{n=1}^{15}\omega^{2n} + \sum_{n=1}^{15}\frac{1}{\omega^{2n}} + 2(-1)\right|$$

$$\left|\frac{\omega^2\left(\omega^{30} - 1\right)}{\left(\omega^2 - 1\right)} + \frac{\frac{1}{\omega^2}\left(\frac{1}{\omega^{30}} - 1\right)}{\left(\frac{1}{\omega^2} - 1\right)} - 2\right|$$

$$\left|\frac{\omega^2\left(\omega^{30} - 1\right)}{\left(\omega^2 - 1\right)} + \frac{\omega^{30} - 1}{\omega^2\left(\omega^2 - 1\right)} - 2\right|$$

$$\left|\frac{\omega^2(1-1)}{\left(\omega^2 - 1\right)} + \frac{1-1}{\omega^2\left(\omega^2 - 1\right)} - 2\right|$$

$= |0 + 0 - 2| = 2$

31. The number of elements in the set $\{z = a + ib \in C : a, b \in Z \text{ and } 1 < |z - 3 + 2i| < 4\}$ is _____.

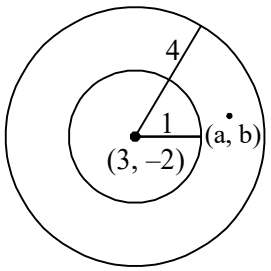
[JEE(Main)June2022]

समुच्चय $\{z = a + ib \in C : a, b \in Z$ तथा $1 < |z - 3 + 2i| < 4\}$ हैं में अवयवों की संख्या है_____.

[JEE(Main)June2022]

Ans. 40

Sol. $1< z - 3 + 2i < 4$



$1 < (a - 3)^2 + (b + 2)^2 < 16$

$(0, \pm2)$, $(\pm2, 0)$, $(\pm1, \pm2)$, $(\pm2, \pm1)$, $(\pm2, \pm3)$, $(3\pm, \pm2)$, $(\pm1, \pm1)$, $(2\pm, \pm2)$, $(\pm3, 0)$, $(0, \pm3)$, $(\pm3, \pm1)$, $(\pm1, \pm3)$,

The total 40 points

32. Sum of squares of modulus of all the complex numbers z satisfying $\overline{z} = iz^2 + z^2 - z$ is equal to _____.

[JEE(Main)June2022]

$\overline{z} = iz^2 + z^2 - z$ का संतुष्ट करने वाली सभी सम्मिश्र संख्याओं z के मापाकों के वर्गों का योगफल बराबर है_____

[JEE(Main)June2022]

Ans. 2

Sol. Let $z = x + iy$ given $\overline{z} = i\left(z^2\right) + z^2 - z$

So $2x = (1+i)\left(x^2 - y^2 + 2ixy\right)$

Separating real and imaginary parts

$\Rightarrow 2x = x^2 - y^2 + 2xy$ .....(1)

$x^2 - y^2 - 2xy = 0$ .....(2)

from eqn. (1) and (2)

$$x = 0 \text{ or } y = -\frac{1}{2}$$

When $x = 0$ $\Rightarrow y = 0$

When $y = -\frac{1}{2}$ $\Rightarrow x^2 - x - \frac{1}{4} = 0$

$$x = \frac{-1 \pm \sqrt{2}}{2}$$

So there will be 3 complex numbers.

$(0+0i);\frac{-1-\sqrt{2}}{2}+i\left(-\frac{1}{2}\right);$ $\frac{-1+\sqrt{2}}{2}+\left(-\frac{i}{2}\right)$

Sum of squares of modulus of these

$$=0+\left(\frac{\sqrt{2}-1}{2}\right)^2+\frac{1}{4}+\left(\frac{\sqrt{2}+1}{2}\right)^2+\frac{1}{4}$$

$$=\frac{3}{2}+\frac{1}{2}=2$$

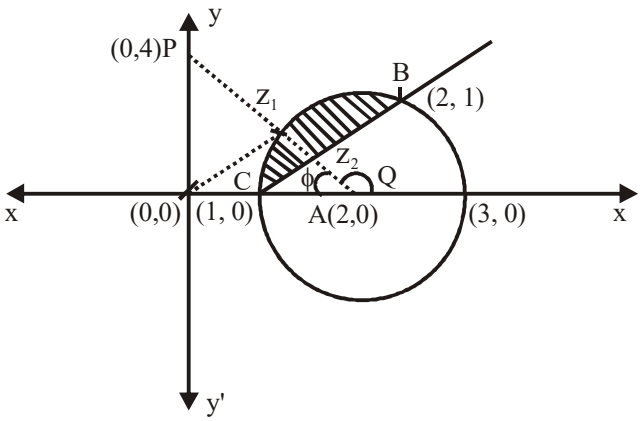
33. Let $S=\{z\in C:|z-2|\le 1, z(1+i)+\bar{z}(1-i)\le 2\}$. Let $|z-4i|$ attains minimum and maximum values, respectively, at $z_1\in S$ and $z_2\in S$. If $5(|z_1|^2+|z_2|^2)=\alpha+\beta\sqrt{5}$, where α and β are integers, then the value of $\alpha+\beta$ is equal to _____. **[JEE(Main)June2022]**

माना $S=\{z\in C:|z-2|\le 1, z(1+i)+\bar{z}(1-i)\le 2\}$ है। माना $|z-4i|$ के निम्नतम तथा उच्चतम मान क्रमशः $z_1\in S$ तथा $z_2\in S$ पर प्राप्त होते हैं। यदि $5(|z_1|^2+|z_2|^2)=\alpha+\beta\sqrt{5}$ है, जहाँ α तथा β पूर्णांक हैं, तो $\alpha+\beta$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. 26

Sol. $S=\{z\in C:|z-2|\le 1, z(1+i)+\bar{z}(1-i)\le 2\}$

S represent shaded region



S represents the shaded region shown in the diagram.

Clearly z_1 will be the point of intersection PA and circle.

PA : $2x + y = 4$

equation of circle $(x-2)^2+y^2=1$

on solving we get $z_1=\left(2-\frac{1}{\sqrt{5}}\right)+\frac{2}{\sqrt{5}}i$

$|z_1|^2=5-\frac{4\sqrt{5}}{5}$

z_2 will will be either B or C

$PB=\sqrt{17}$ and $PC=\sqrt{13}$

Hence $z_2=1$

$5\left(|z_1|^2+|z_2|^2\right)=30-4\sqrt{5}$

Clearly $\alpha = 30$ and $\alpha = -4 \Rightarrow \alpha+\beta=26$

34. If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4+x^3+x^2+x+1=0$, then $\alpha^{2021}+\beta^{2021}+\gamma^{2021}+\delta^{2021}$ is equal to :

[JEE(Main)July2022]

यदि समीकरण $x^4+x^3+x^2+x+1=0$ के मूल $\alpha, \beta, \gamma, \delta$ हैं, तो $\alpha^{2021}+\beta^{2021}+\gamma^{2021}+\delta^{2021}$ बराबर हैं: :

[JEE(Main)July2022]

(A) –4 (B) –1 (C) 1 (D) 4

Ans. B

Sol. $x^4+x^3+x^2+x+1=0$ OR $\frac{x^5-1}{x-1}=0(x\neq 1)$

So roots are $e^{i2\pi/5}, e^{i4\pi/5}, e^{i6\pi/5}, e^{i8\pi/5}$

i.e. α, β, γ and δ

From properties of nth root of unity

$1^{2021}+\alpha^{2021}+\beta^{2021}+\gamma^{2021}+\delta^{2021}=0$

$\Rightarrow \alpha^{2021}+\beta^{2021}+\gamma^{2021}+\delta^{2021}=-1$

35. For $n \in N$, let $S_n=\left\{z\in C:|z-3+2i|=\frac{n}{4}\right\}$ and $T_n=\left\{z\in C:|z-2+3i|=\frac{1}{n}\right\}$. Then the number of elements in the set $\{n \in N : S_n \cap T_n=\phi\}$ is : **[JEE(Main)July2022]**

$n \in N$ के लिए, $S_n=\left\{z\in C:|z-3+2i|=\frac{n}{4}\right\}$ तथा $T_n=\left\{z\in C:|z-2+3i|=\frac{1}{n}\right\}$ हैं। तो समुच्चय $\{n \in N : S_n \cap T_n=\phi\}$ में अवयवों की संख्या है : **[JEE(Main)July2022]**

(A) 0 (B) 2 (C) 3 (D) 4

Ans. D

Sol. $S_n : |z-(3-2i)|=\frac{n}{4}$ is a circle center $C_1(3,-2)$ and radius n/4

$T_n : |Z-(2-3i)| = \frac{1}{n}$ is a circle center $C_2(2,-3)$ and radius $1/n$

Here $S_n \cap T_n = \phi$

Both circles do not intersec each other

Case-1 : $C_1C_2 > n/4 + 1/n$

$$\sqrt{2} > \frac{n}{4} + \frac{1}{n}$$

then n = 1, 2, 3, 4

Case-2 : $C_1C_2 < \left|\frac{n}{4} - \frac{1}{n}\right|$

$$\Rightarrow \sqrt{2} < \left|\frac{n^2-4}{4n}\right|$$

$\Rightarrow$ n has infinite solutions for $n \in \mathbf{N}$

36. For $z \in C$ if the minimum value of $\left(\left|z-3\sqrt{2}\right| + \left|z-p\sqrt{2}i\right|\right)$ is $5\sqrt{2}$, then a value of p is :

[JEE(Main)July2022]

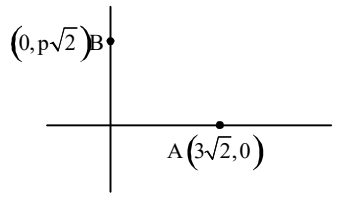
$z \in C$, के लिए यदि $\left(\left|z-3\sqrt{2}\right| + \left|z-p\sqrt{2}i\right|\right)$ का निम्नतम मान $5\sqrt{2}$ है, तो p का एक मान है :

[JEE(Main)July2022]

(A) 3 (B) $\frac{7}{2}$ (C) 4 (D) $\frac{9}{2}$

Ans. C

Sol.



It is sum of distance of z from $(3\sqrt{2}, 0)$ and $(0, p\sqrt{2})$

For minimising, z should lie on AB and AB = $5\sqrt{2}$

$(AB)^2 = 18 + 2p^2$

$p = \pm 4$

37. Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $Re(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

[JEE(Main)July2022]

माना O मूल बिन्दु है तथा $z_1 = 1 + 2i$ है। यदि B, बिन्दु z_2, $Re(z_2) < 0$, है तथा OAB एक समद्विबाहु समकोण त्रिभुज है, जिसका कर्ण OB है, तो निम्न में से कौनसा सत्य नहीं है ?

[JEE(Main)July2022]

(A) $\arg z_2 = \pi - \tan^{-1} 3$

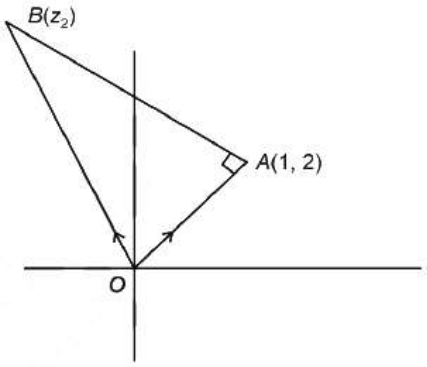
(B) $\arg (z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$

(C) $|z_2| = \sqrt{10}$

(D) $|2z_1 - z_2| = 5$

Ans. D

Sol.



$$\frac{z_2 - 0}{(1+2i) - 0} = \frac{|OB|}{|OA|} e^{\frac{i\pi}{4}}$$

$$\Rightarrow \frac{z_2}{1+2i} = \sqrt{2} e^{\frac{i\pi}{4}}$$

Or $z_2 = (1 + 2i)(1 + i)$

$= -1 + 3i$

$\arg z_2 = \pi - \tan^{-1} 3$

$|z_2| = \sqrt{10}$

$z_1 - 2z_2 = (1 + 2i) + 2 - 6i = 3 - 4i$

$$\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$$

$|2z_1 - z_2| = |2 + 4i + 1 - 3i| = |3 + i|$

$= \sqrt{10}$

38. If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then : **[JEE(Main)July2022]**

यदि $z = x + iy$ समीकरणों $|z| - 2 = 0$ तथा $|z - i| - |z + 5i| = 0$ को संतुष्ट करता है, तो : **[JEE(Main)July2022]**

(A) $x + 2y - 4 = 0$ (B) $x^2 + y - 4 = 0$

(C) $x + 2y + 4 = 0$ (D) $x^2 - y + 3 = 0$

Ans. C

Sol. $|z-i|-|z+5i|=0$

$\Rightarrow |x + (y - 1)i| = |x + (y + 5)i|$

$x^2 + (y - 1)^2 = x^2 + (y + 5)^2$

$(y - 1)2 - (y + 5)^2 = 0$

$(2y + 4)(-6) = 0$

$y = -2$

$\therefore x^2 + (-2)^2 = 4$

$x = 0$

$Z \equiv (0, -2)$, check options

39. Let the minimum value v_0 of $v = |z|^2 + |z - 3|^2 + |z - 6i|^2, z \in C$ is attained at $z = z_0$. Then $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$ is equal to : **[JEE(Main)July2022]**

माना v_0 of $v = |z|^2 + |z - 3|^2 + |z - 6i|^2$, $z \in C$ का न्यूनतम मान $z = z_0$ पर प्राप्त होता है। तो $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$ बराबर है : **[JEE(Main)July2022]**

(A) 1000 (B) 1024 (C) 1105 (D) 1196

Ans. A

Sol. $z_0 = \left(\frac{0+3+0}{3}, \frac{0+6+0}{3}\right) = (1, 2)$

$v_0 = |1 + 2i|^2 + |1 + 2i - 3|^2 + |1 + 2i + 6i|^2 = 30$

Then $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$

$= \left|2(1 + 2i)^2 - (1 - 2i)^3 + 3\right|^2 + 900$

$= \left|2(1 - 4 + 4i) - (1 - 4 - 4i)(1 - 2i) + 3\right|^2 + 900$

$= |8 + 6i|^2 + 900 = 100 + 900 = 1000$

40. Let S be the set of all (α, β), $\pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta$, $(\alpha, \beta) \in S$. Then $\sum_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta} + \frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to:

[JEE(Main)July2022]

माना सभी (α, β), $\pi < \alpha, \beta < 2\pi$, जिनके लिए सम्मिश्र संख्या $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ विशुद्ध काल्पनिक तथा $\frac{1+i\cos\beta}{1-2i\cos\beta}$ विशुद्ध वास्तविक है, का समुच्चय S है। माना $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta$, $(\alpha, \beta) \in S$ हैं। तो $\sum_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta} + \frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ बराबर है :

[JEE(Main)July2022]

(A) 3 (B) $3i$ (C) 1 (D) $2 - i$

Ans. C

Sol. $\pi < \alpha, \beta < 2\pi$

$$\frac{1-i\sin\alpha}{1+i(2\sin\alpha)} = \text{Purely imaginary}$$

$$\Rightarrow \frac{(1-i\sin\alpha)(1-i(2\sin\alpha))}{1+4\sin^2\alpha} = \text{Purely imaginary}$$

$$\Rightarrow \frac{1-2\sin^2\alpha}{1+4\sin^2\alpha} = 0$$

$$\Rightarrow \sin^2\alpha = \frac{1}{2}$$

$$\Rightarrow \alpha = \left\{\frac{5\pi}{4}, \frac{7\pi}{4}\right\}$$

$$\& \ \frac{1+i\cos\beta}{1+i(-2\cos\beta)} = \text{Purely real}$$

$$\Rightarrow \frac{(1+i\cos\beta)(1+2i\cos\beta)}{1+4\cos^2\beta} = \text{Purely real}$$

$$\Rightarrow 3\cos\beta = 0$$

$$\Rightarrow \beta = \frac{3\pi}{2}$$

$$\Rightarrow z_{\alpha\beta} = \sin\frac{5\pi}{2} + i\cos 3\pi = 1 - i$$

or

$$z_{\alpha\beta} = \sin\frac{7\pi}{2} + i\cos 3\pi = -1 - i$$

$$\text{Required value} = \left[i(1-i) + \frac{1}{i(1+i)}\right]\left[i(-1-i) + \frac{1}{i(-1+i)}\right]$$

$$= i(-2i) + \frac{1}{i}\frac{2i}{i(-2)} \Rightarrow 2 - 1 = 1$$

41. Let $S_1 = \left\{z_1 \in C : |z_1 - 3| = \frac{1}{2}\right\}$ and $S_2 = \left\{z_2 \in C : \left|z_2 - |z_2 + 1|\right| = \left|z_2 + |z_2 - 1|\right|\right\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2 - z_1|$ is : **[JEE(Main)July2022]**

माना $S_1 = \left\{z_1 \in C : |z_1 - 3| = \frac{1}{2}\right\}$ तथा $S_2 = \left\{z_2 \in C : \left|z_2 - |z_2 + 1|\right| = \left|z_2 + |z_2 - 1|\right|\right\}$ हैं। तो $z_1 \in S_1$ तथा $z_2 \in S_2$ के लिए $|z_2 - z_1|$ का निम्नतम मान है : **[JEE(Main)July2022]**

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Ans. C

Sol. $\left|z_2 + |z_2 - 1|\right|^2 = \left|z_2 - |z_2 + 1|\right|^2$

$$\Rightarrow \left|z_2 + |z_2 - 1|\right|\left(\bar{z}_2 + |z_2 - 1|\right) = \left(z_2 - |z_2 + 1|\right)\left(\bar{z}_2 - (z_2 + 1)\right)$$

$$\Rightarrow z_2\left|\bar{z}_2 + 12_2 - 1\right| - \left(\bar{z}_2 - |z_2 + 1|\right) + \bar{z}_2\left(|z_2 - 1| + |z_2 + 1|\right)$$

$$= |z_2 + 1|^2 = |z_2 - 1|^2$$

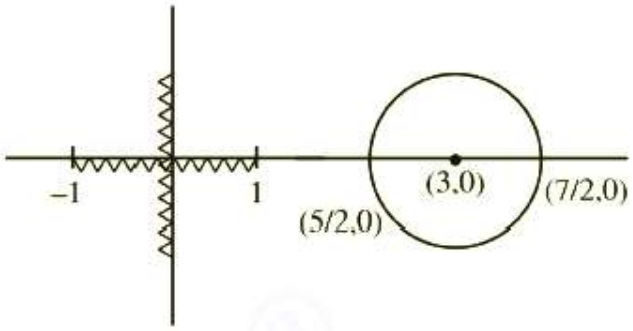
$$\Rightarrow \left[z_2 + \bar{z}_2\right)\left(|z_2 - 1|\right) + (z_2 + 1) = 2\left(z_2 + \bar{z}_2\right)$$

$$\Rightarrow \left(z_2 + \bar{z}_2\right)\left(|z_2 - 1| + |z_2 + 1| - 2\right) = 0$$

$\therefore z_2 + \bar{z}_2 = 0$ or $|z_2 - 1| + |z_2 + 1| - 2 = 0$

$\therefore z_2$ lie on imaginary axis. Or on real axis within [–1, 1]

Also $|z_1 - 3| = \frac{1}{2}$ lie on circle having centre 3 and radius $\frac{1}{2}$.



Clearly $|z_1 - z_2|\ \min = \frac{5}{2} - 1 = \frac{3}{2}$

42. If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to : **[JEE(Main)July2022]**

यदि $z = 2 + 3i$ है, तो $z^5 + (\bar{z})^5$ बराबर है : **[JEE(Main)July2022]**

(A) 244 (B) 224 (C) 245 (D) 265

Ans. A

Sol. $z^5 + (\bar{z})^5 = (2+3i)^5 + (2-3i)^5$

$= 2\left({}^5C_0 2^5 + {}^5C_2 2^3 (3i)^2 + {}^5C_4 2^1 (3i)^4 \right)$

$= 2\,(32 + 10 \times 8\,(-9) + 5 \times 2 \times 81) = 244$

43. If $z \neq 0$ be a complex number such that $\left|z - \frac{1}{z}\right| = 2$, then the maximum value of $|z|$ is :

[JEE(Main)July2022]

यदि सम्मिश्र संख्या $z \neq 0$ के लिए $\left|z - \frac{1}{z}\right| = 2$ है, तो $|z|$ का अधिकतम मान है : **[JEE(Main)July2022]**

(A) $\sqrt{2}$ (B) 1 (C) $\sqrt{2} - 1$ (D) $\sqrt{2} + 1$

Ans. D

Sol. $z - 1/z| = 2$

$|z| - \frac{1}{|z|} \leq |z - \frac{1}{z}| \leq |z| + \frac{1}{|z|}$ Let $|z| = r$

$\left|r - \frac{1}{r}\right| \leq 2 \leq r + \frac{1}{r}$

$\left|r-\frac{1}{r}\right| \le 2$ & $r+\frac{1}{r} \ge 2$ always true

$r-\frac{1}{r} \ge -2$ & $r-\frac{1}{r} \le 2$

$r^2-1 \le 2r$

$r^2-2r \le 1$

$(r-1)^2 \le 2$

$r-1 \le \sqrt{2}$

$\therefore |z|_{max} = 1+\sqrt{2}$

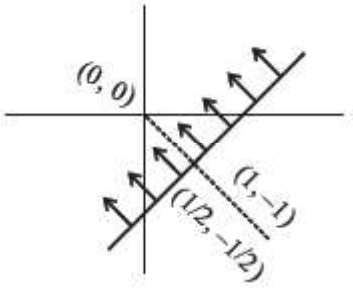
44. Let $S = \{z = x + iy : |z - 1 + i| \ge |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of x, for which $w = 2x + iy \in S$ for some $y \in R$, is : **[JEE(Main)July2022]**

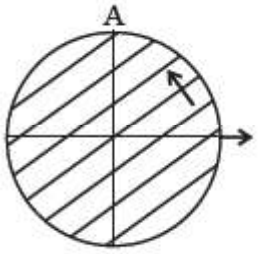
माना $S = \{z = x + iy : |z - 1 + i| \ge |z|, |z| < 2, |z + i| = |z - 1|\}$ है। तो x के उन सभी मानों, जिसके लिए किसी $y \in R$ के लिए $w = 2x + iy \in S$ है, का समुच्चय है : **[JEE(Main)July2022]**

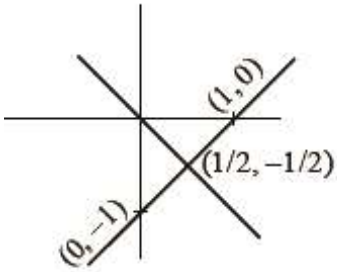
(A) $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$ (B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$ (C) $\left(-\sqrt{2}, \frac{1}{2}\right]$ (D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

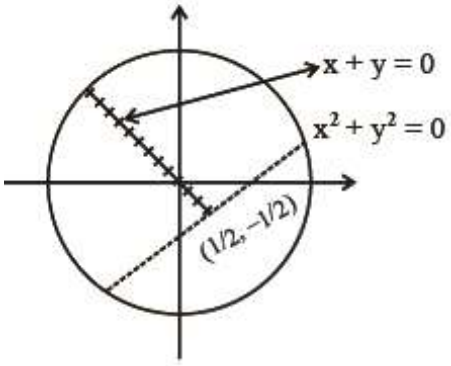
Ans. B

Sol. $|z-1+i| \ge |z|$; $|z| < 2$; $|z+i| = |z-1|$









Hence

$w = 2x + iy \in S$

$2x \le \frac{1}{2} \quad \therefore x \le \frac{1}{4}$

Now

$(2x)^2 + (2x)^2 < 4$

$x^2 < \frac{1}{2} \Rightarrow x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

$\therefore x \in \left(\frac{-1}{\sqrt{2}}, \frac{1}{4}\right]$

45. Let $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ and B = A – I. If $\omega = \frac{\sqrt{3}i - 1}{2}$, then the number of elements in the set $\left\{n \in \{1, 2, ..., 100\} : A^n + (\omega B)^n = A + B\right\}$ is equal to _____. **[JEE(Main)July2022]**

माना $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ तथा $B = A - I$ हैं। यदि $\omega = \frac{\sqrt{3}i - 1}{2}$ है, तो $\left\{n \in \{1, 2, ..., 100\} : A^n + (\omega B)^n = A + B\right\}$ है, में अवयवों की संख्या है _____ **[JEE(Main)July2022]**

Ans. 17

Sol. $A=\begin{bmatrix}2 & -1 & -1\\ 1 & 0 & -1\\ 1 & -1 & 0\end{bmatrix} \Rightarrow A^2 = A \Rightarrow A^n = A.$

$\forall n \in \{1,2,.....,100\}$

Now, $B = A - I = \begin{bmatrix}1 & -1 & -1\\ 1 & -1 & -1\\ 1 & -1 & -1\end{bmatrix}$

$B^2 = -B$

$\Rightarrow B^3 = -B^2 = B$

$\Rightarrow B^5 = B$

$\Rightarrow B^{99} = B$

Also, $\omega^{3k} = 1$

So, n = common of {1, 3, 5, ., 99} and

{3, 6, 9, ., 99} = 17

46. Let $S=\{z \in C : z^2 + \overline{z} = 0\}$. Then $\sum_{z \in S}(Re(z) + Im(z))$ is equal to _____.

[JEE(Main)July2022]

माना $S=\{z \in C : z^2 + \overline{z} = 0\}$ है, तो $\sum_{z \in S}(Re(z) + Im(z))$ बराबर है _____ **[JEE(Main)July2022]**

Ans. 0

Sol. $S = \{z \in C : z^2 + \overline{z} = 0\}$

Let $z = x + iy$

$z^2 = x^2 - y^2 + 2ixy$

$\overline{z} = x - iy$

$z^2 + \overline{z} = x^2 - y^2 + x + i(2xy - y) = 0$

$\Rightarrow x^2 + x - y^2 = 0$ & $2xy - y = 0$

$y = 0$ or $x = \frac{1}{2}$

If $y = 0$; $x = 0, -1$

If $x = \frac{1}{2}; y = \frac{\sqrt{3}}{2}, \frac{-\sqrt{3}}{2}$

$$\sum_{z\in S}\left(\operatorname{Re}(z)+\operatorname{Im}(z)=\left(0-1+\frac{1}{2}+\frac{1}{2}\right)+0+0\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}\right)$$

47. Let $z = a + ib$, $b \neq 0$ be complex numbers satisfying $z^2 = \bar{z}.2^{1-|z|}$. Then the least value of $n \in N$, such that $z^n = (z + 1)^n$, is equal to _____ **[JEE(Main)July2022]**

माना सम्मिश्र संख्याऐं $z = a + ib$, $b \neq 0$ समीकरण $z^2 = \bar{z}.2^{1-|z|}$ को संतुष्ट करती हैं। तो $n \in N$ का निम्नतम मान, जिसके लिए $z^n = (z + 1)^n$ है, बराबर है _____ **[JEE(Main)July2022]**

Ans. 6

Sol. $\left|z^2\right| = \left|\bar{z}\right|.2^{1-|z|} \Rightarrow |z| = 1$

$z^2 = \bar{z} \Rightarrow z^3 = 1 \therefore z = \omega \text{ or } \omega^2$

$\omega^n = (1+\omega)^n = \left(-\omega^2\right)^n$

Least natural value of n is 6.